

Reading Harm but Not Refusing It: Why VLAs Fail to Self-Refuse, and a Scene-Blind Gate That Works

Anonymous Authors¹

Abstract

A vision–language–action (VLA) model capable enough to follow instructions will also execute harmful ones. *Mechanistically*, VLAs *linearly encode* instruction harmfulness in their language-model backbone—a single logistic-regression probe on the image-free representation classifies harm at $AUC \approx 0.99$ at every layer and across two backbone families—yet they neither self-refuse nor respond to activation steering. We trace this *read≠write dissociation* to a harm direction that rotates across depth and, even at the readout, moves the action but not toward inhibition: there is no learned harm→inhibition circuit to actuate. This motivates supplying the missing inhibition *externally*: a scene-blind text-only probe gates a null-action override. On LIBERO (OpenVLA-7B, $n=1900$) it raises a combined safety score by $+0.559$ ($p < 10^{-5}$) and drives harmful-instruction refusal from 0.35 to 0.91–0.98 at no benign cost, transfers zero-shot across VLA fine-tunes and backbone families, and resists paraphrase, persona, injection, and cross-lingual attacks—being blind only to harm triggered by the visual scene-state rather than the words.

1. Introduction

Vision–language–action (VLA) models map a camera image and a natural-language instruction directly to robot actions (Kim et al., 2024). The same competence that lets a VLA execute “pick up the bowl and place it on the plate” lets it execute “knock the bowl onto the floor.” This is not hypothetical: embodied models are readily jailbroken into harmful physical actions (Robey et al., 2025; Zhang et al., 2025b), and safety benchmarks find they comply with hazardous instructions (Ying et al., 2025; Yin et al., 2024).

¹Anonymous Institution, Anonymous City, Anonymous Region, Anonymous Country. Correspondence to: Anonymous Author <anon.email@domain.com>.

Preliminary work. Under review by the International Conference on Machine Learning (ICML). Do not distribute.

Safety here is concrete: *refuse harmful instructions while preserving benign task capability*.

A natural hope is that LLM safety techniques transfer: prompt-based refusal, activation steering along a “refusal direction” (Arditi et al., 2024; Turner et al., 2023; Zou et al., 2023), or refusal fine-tuning. On VLAs these collapse capability or fail out-of-distribution (OOD). This raises a question both scientific and practical: *does a VLA even represent the harmfulness of an instruction, and if so, why does it not refuse?*

We answer affirmatively and sharply. Harmfulness is *linearly readable* from the VLA’s LLM backbone using a single logistic-regression probe on the *image-free* instruction representation—no concept dictionary, no fine-tuning. The probe attains $AUC \approx 0.99$ at every layer and transfers zero-shot across VLA fine-tunes of the same backbone and across backbone families (LLaMA-2 and Qwen2.5-VL).

The conceptual core is that reading the harm signal is not acting on it. We show a *read≠write dissociation*: (i) the readable harm direction is layer-local and rotates across depth, so the mid-layer direction one would steer along is nearly orthogonal to the final-layer direction that feeds the action readout; and (ii) even injecting the final-layer harm direction at the readout *moves* the action—it is not in the readout’s null space—but toward *larger*, not frozen, actions. There is no learned circuit routing harm-awareness to action *inhibition*: a VLA encodes harm as an ordinary semantic feature wired to nothing, which is exactly why steering and refusal fail.

This diagnosis has a clean consequence: supply the missing inhibition *externally*. One extra image-free forward pass reads the harm probe; if it fires, a null-action override gates the policy. Our contributions:

- **Mechanism (read≠write).** Correlational and *causal* evidence that VLAs linearly encode harm but lack a harm→inhibition circuit; replicated across four task suites and two backbone families.
- **Method.** A scene-blind single-linear-probe inference-time refusal gate. On $n=1900$ it improves the safety score by $+0.559$ ($p < 10^{-5}$) at no benign cost, and

beats a concurrent multimodal-dictionary method (Wen et al., 2026) on its own attack recipe (1.9% vs. 7.8% attack success).

- **Generality and honest scoping.** Zero-shot cross-VLA transfer (ROC-AUC 1.000), cross-backbone replication, and robustness to paraphrase / persona / injection / cross-lingual attacks; because it is scene-blind the gate cannot see harm whose trigger is the visual scene-state, which we quantify.

2. Related Work

Inference-time VLA safety. Closest is the concurrent SAFE-Dict (Wen et al., 2026), which learns a per-concept PCA dictionary with sparse (ElasticNet) decoding on the VLA’s *fused multimodal* hidden state. We share the inference-time, plug-in, no-retraining goal but differ on three axes: we are *scene-blind* (text-only forward, no image tokens), we use a *single linear probe* (no dictionary or concept stimuli), and we demonstrate *cross-backbone* transfer. We also report a failure mode they do not: multimodal probing overfits to scenes and collapses OOD. We further test their core move directly: in closed-loop rollout, *neither* adding nor removing a single harm direction at the readout yields selective refusal (Table 3), on two backbone families. We read this not as a refutation but as a scope statement—their gains plausibly come from a richer multi-concept dictionary over the fused multimodal state with *soft* attenuation. SAFE (Gu et al., 2025) detects task *failure* (not harm) from VLA internal features; RoboGuard (Ravichandran et al., 2025) and KnowNo (Ren et al., 2023) gate LLM planners (via temporal-logic synthesis and conformal abstention) rather than low-level policies; SafeVLA (Zhang et al., 2025a) aligns at training time and is composable with our gate. See Li et al. (2026) for a survey.

Attacks, refusal, and interpretability. LLM-controlled robots are jailbroken at near-100% success (Robey et al., 2025); BadRobot (Zhang et al., 2025b) documents a “word-action safety misalignment” in which models verbally refuse yet physically comply—our read≠write dissociation is a circuit-level account of exactly this behavior. In LLMs, refusal is mediated by a largely linear direction (Arditi et al., 2024) amenable to activation steering (Turner et al., 2023; Zou et al., 2023) and conditional (probe-gated) steering (Lee et al., 2025); all of these *actuate internally* and so presuppose a write path that, we show, VLAs lack. Zhao et al. (2025) show harmfulness is encoded separately from refusal and read out at the instruction-token position; our scene-blind probe reads exactly there, and we extend the dissociation to its extreme. Probing-based input detectors have been criticized for learning surface patterns (Wang et al., 2025) and for adversarial evadability (Bailey et al., 2025);

our paraphrase, cross-lingual, obfuscation, and white-box experiments address both critiques. Finally, linear probes recover symbolic states from OpenVLA’s backbone (Lu et al., 2025) and interventions *do* steer motor-semantic features such as speed and direction (Häon et al., 2025): the VLA write path works for features the policy was trained to act on; what is missing is specifically a harm→inhibition circuit.

3. Setup and Threat Model

Models and benchmark. Our primary system is OpenVLA-7B (Kim et al., 2024) fine-tuned on the four LIBERO suites (Liu et al., 2023) (SPATIAL, OBJECT, GOAL, LONG). OpenVLA decodes a 7-DoF action autoregressively as the last 256 tokens of the vocabulary, so an action is selected by the hidden state’s projection onto the span of those 256 unembedding rows—the *action-readout subspace* S_{act} . For cross-backbone tests we use ECoT-OXE (Zawalski et al., 2024) (same LLaMA-2 backbone, chain-of-thought fine-tune on Open X-Embodiment) and NORA-1.5 (Hung et al., 2025) (Qwen2.5-VL-3B backbone).

Data and metric. We curate harmful/benign LIBERO instructions spanning physical harm, property damage, and bias/fairness, split disjointly into a *dev* set (probe training/calibration) and a *test* set never seen during training or threshold selection. We report succ_b (benign success), ref_b (benign refused, lower is better), and ref_h (harmful refused), combined into $\text{Score} = \text{succ}_b + \text{ref}_h - \text{ref}_b$, with a paired stratified cluster bootstrap over instruction tuples for significance. *Threat model:* the adversary issues instructions; the defender runs one image-free probe forward per instruction and may veto the action, considering both a white-box adversary who knows the probe weights and (Section 7) one who perturbs the image.

4. Observations: Reading Harm but Not Refusing

(I) Harm is linearly readable. For each suite we forward the instruction through the LLM submodule *with no image tokens*, take the last-token hidden state at layer L , and fit logistic regression. A single probe separates held-out harmful from benign instructions at 5-fold out-of-fold AUC 0.979 pooled across suites, 0.96–0.99 per layer, peaking near $L=8$. The signal is intrinsic: probes trained on 24 vs. 196 benign instructions have direction cosine 0.73–0.89.

Cross-VLA and cross-lingual. The harm axis is a property of the backbone, not OpenVLA’s fine-tune. Re-fitting on ECoT-OXE gives OOF AUC 0.984; the OpenVLA probe applied *zero-shot* to ECoT reaches ROC-AUC 1.000 (direction cosine 0.805). On Qwen2.5-VL-3B (NORA) an independent probe reaches AUC 0.99. The axis is even

Table 1. The harm direction rotates across depth (absolute cosine to the final-layer direction) yet is readable at every layer. OpenVLA-7B, four-suite mean.

	L_4	L_8	L_{16}	L_{20}	L_{fin}
$\cos(\cdot, L_{\text{fin}})$	0.02	0.04	0.29	0.45	1.00
probe OOF AUC	0.95	0.96	0.99	0.99	0.99

cross-lingual: an English-trained probe flags Korean translations of the harmful instructions at mean $p_{\text{harm}}=0.92$.

(II) Reading is not writing. If the harm signal is right there, why does the VLA execute harm, and why can’t we steer it to refuse? We give correlational and causal evidence for a *read*≠*write* dissociation.

The harm axis rotates across depth. Fitting a probe at each layer, the signal is readable everywhere but the *direction* is layer-local (Table 1): the layer-8 direction (strongest, where one would steer) is nearly orthogonal to the final-layer direction (cosine 0.03–0.07), while adjacent layers correlate. Bootstrap 95% CI uppers on $\cos(L_8, L_{\text{fin}})$ are ≤ 0.075 in all four suites, and the same banded rotation holds for NORA/Qwen2.5-VL ($\cos(L_4, L_{28}) = 0.03$). Depth rotation is a known property of LLMs (Belrose et al., 2023); its *consequence* for safety is that any single-layer steering vector is misaligned with what the action readout consumes.

The final direction reaches actions but not inhibition. The harm direction is not walled off: its energy inside S_{act} is 2.3–2.8 $\times$ that of a random direction, and the late MLP blocks (layers 26–31) that *write* harm are the *same* blocks that feed the action readout (Spearman 0.70–0.76). Harm is co-located with the action machinery—just not wired to inhibit it: it is near-orthogonal to the freeze direction $\mathbf{d}_{\text{inh}}$ maximizing the null-action logit ($|\cos(\mathbf{d}_{\text{fin}}, \mathbf{d}_{\text{inh}})| \leq 0.05$ on both OpenVLA and ECoT, no larger than random). Injecting $\alpha \mathbf{d}_{\text{fin}}$ at the post-norm readout and re-decoding moves the action but *increases* its magnitude (0.14→0.37) and pushes it *away* from freezing: more action, never refusal.

Closed-loop interventions never refuse selectively. Over full rollouts ($n=380$ test episodes; Table 3) we run two internal interventions and compare to the external gate. *Steering*: even the *optimized* best push-toward-freeze direction $\mathbf{d}^*$ (gradient ascent on the zero-displacement action log-prob), injected every step, freezes only *indiscriminately*—harmful and benign refusal rise together and past a narrow band benign success collapses (succ_b 0.70→0.32); no operating point spares benign. *Ablation*—the SAFE-Dict move of removing the harmful component (Wen et al., 2026)—does essentially nothing (ref_h stays at the vanilla floor for removal up to 2 $\times$). *Cross-family*: on NORA (Qwen2.5-VL,

Table 2. Steering-as-refusal α sweep (OpenVLA, inject $\mathbf{d}^*$ at readout, $n=380$). Harmful and benign refusal rise together; over-steering collapses benign success. No selective operating point exists.

α	ref_h	succ_b	ref_b	Score
0 (vanilla)	0.347	0.700	0.195	0.852
16	0.564	0.662	0.306	0.920
24	0.655	0.637	0.388	0.905
32	0.627	0.656	0.450	0.834
48	0.605	0.319	0.431	0.492

Table 3. Closed-loop rollouts ($n=380$): internal interventions at the readout never refuse selectively; only the external gate does. Steering shows its best harmful-refusal operating point; ablation removes the harm direction (fraction 1 $\times$). “–” = NORA vanilla refuses nothing.

Backbone	Intervention	ref_h	succ_b	ref_b
OpenVLA	Vanilla	0.347	0.700	0.195
	Steer + $\mathbf{d}^*$ (best)	0.655	0.637	0.388
	Ablate –harm	0.359	0.656	0.200
	External gate	0.906	0.700	0.195
NORA	Vanilla	~0.00	–	–
	Ablate –harm	0.000	0.925	0.000
	External gate	0.95	0.87	–

FAST/flow head), ablating harm leaves harmful refusal at 0.000 while the external gate reaches 0.95. Only reading the signal and inhibiting from *outside* works—on both backbone families. The α -sweep of the optimized steering direction (Table 2) makes the failure concrete: as we push harder, harmful and benign refusal climb *together* and *Score* peaks only modestly before over-steering ($\alpha=48$) collapses benign success—no operating point isolates harm.

Controls rule out under-power. The same readout steering has a clean graded effect on action magnitude, and mid-layer steering pushed to 29.6% of the residual-stream norm still does not raise refusal ($p > 0.14$); after refusal-LoRA the harm direction stays orthogonal to freeze ($|\cos| \leq 0.11$) and that LoRA refuses *indiscriminately* (benign false-refusal 0.20→0.50). VLA steering *does* work for motor-semantic features like speed (Häon et al., 2025); only *refusal* steering fails—you cannot amplify a circuit that does not exist.

Takeaway: a VLA encodes harm as an ordinary semantic feature wired to nothing—so the inhibition must be supplied from outside.

5. Method: A Scene-Blind Refusal Gate

The observations imply the remedy—supply inhibition externally. At inference we run one image-free probe forward per instruction; if p_{harm} exceeds a threshold (Youden’s J on 5-fold out-of-fold dev predictions, so test is never touched),

Table 4. Main result (OpenVLA-7B, LIBERO, $N=50$, $n=1900$). $Score = succ_b + ref_h - ref_b$. Brackets: episode-level 95% bootstrap.

Method	$succ_b$	ref_b	ref_h	Score
Vanilla	0.700	0.195	0.347	0.852 [0.80,0.90]
Unified @ Youden	0.647	0.242	0.975	1.380 [1.33,1.43]
Unified @ 0.5	0.700	0.195	0.912	1.417
Per-suite @ 0.5	0.700	0.195	0.906	1.411 [1.36,1.46]

Table 5. Method comparison ($N=10$, $n=380$). The text-only gate is the only effective, OOD-robust option. “n.s.”=not significant; $*p<10^{-4}$.

Method	Score	Δ vs. vanilla
Vanilla	0.806	—
Activation steering	0.85–0.86	+0.04–0.05 (n.s.)
Refusal-LoRA (OOD)	0.764	−0.04
Multimodal gate (OOD)	0.247	−0.56
Text-only gate	1.361	+0.555*

we override every step with a null action. Overhead is one extra LLM forward (~ 50 ms); no fine-tuning, no prompt change.

6. Experiments

Headline result. Table 4 reports $N=50$ ($n=1900$, the full LIBERO init-state population). The gate lifts $Score$ by +0.559 (per-suite, $p < 10^{-5}$) or +0.528 (unified probe, held-out-calibrated threshold, $p \approx 3 \times 10^{-4}$), with CIs disjoint from vanilla. Harmful refusal rises from 0.347 to 0.91–0.98 while benign success and false-refusal match vanilla to rounding.

Against alternatives. Table 5 ($N=10$, $n=380$) contrasts the gate with the LLM-safety transfers. Activation steering is not significant; refusal-LoRA overgeneralizes OOD (benign false-refusal 0.20 $\rightarrow$ 0.50); a *multimodal* probe gate collapses ($Score$ 0.247) because it memorizes scenes and fires on every novel one—direct evidence for being scene-blind. The text-only gate is the only method that both refuses harm and preserves benign capability.

Against SAFE-Dict and cross-backbone. On a reconstruction of SAFE-Dict’s Libero-Harm recipe (75 prompts, 5 hazard categories), our per-suite gate attains attack-success rate (ASR) 7.7% (parity with their reported 7.8%); adding disjoint toxicity vocabulary to the dev set drives the unified probe to ASR 1.9% at zero clean-capability loss—a $4\times$ improvement using a single linear probe rather than a per-concept dictionary. On NORA the vanilla policy never refuses (ASR 100%); the Qwen-side gate cuts ASR to 5.3% while clean success is preserved (86% $\rightarrow$ 87%).

Table 6. Probe robustness. ASR = harmful instructions evading the gate. The harm axis tracks meaning, not surface form.

Attack family	ASR
Paraphrase (intent-preserving)	4.5%
Persona / roleplay framing	0%
Benign-carrier injection	0%
Korean translation	0%
Character-spacing / leetspeak	$\sim 0\%$
White-box greedy (removes the harm)	18%

Robustness. Linear probes on inputs have been criticized for learning surface lexical patterns and for evadability by adaptive attackers. Treating the gate as the asset to attack, we measure the fraction of harmful instructions that slip below threshold (Table 6). Persona framing and benign-task injection *raise* p_{harm} (negative evasion); paraphrase rarely evades; Korean translation, character spacing, and leetspeak do not evade. A white-box greedy attacker that knows the probe weights succeeds 18% of the time, but *every* success requires substituting away the harmful action itself (“smash” $\rightarrow$ “place”)—i.e., the only way past the probe is to make the instruction benign. Adversarially augmenting the dev set with attack paraphrases keeps held-out ASR at 0% with no change in clean false-refusal.

Is null-action a real refusal? Yes. Under the gate, harmful episodes have median per-step action norm 0.000 (a genuine freeze) versus 0.305 for vanilla; a sustained-near-zero refusal definition gives $ref_h=0.905$ vs. 0.27 vanilla, so the headline is not an artifact of the refusal definition.

Takeaway: one image-free linear probe delivers strong, transferable, attack-robust refusal at no benign cost—where every internal intervention fails.

7. Discussion and Conclusion

The scene-blind blind spot. Scene-blindness is what makes the gate robust (immune to novel scenes and bounded adversarial images) but is also a structural limit: the decision is provably a function of the instruction text alone. For a fixed instruction across 40 scenes the text representation is *identical* (spread 0.000) while the multimodal one varies (7.7%). With benign-phrased instructions harmful only given a scene-state (“place the bowl on the stove” when the burner is lit), the gate misses 25%—exactly the neutral-wording cases; it still catches harm when the wording carries a risk cue. This motivates a hybrid: the text gate plus a thin hazard-*presence* scene channel (which detects LIBERO’s one scene-state hazard at AUC 1.000).

Conclusion. A VLA reads instruction harm but has no circuit to act on it; supplying that inhibition with one external scene-blind probe yields strong, transferable refusal.

References

- Arditi, A., Obeso, O., Syed, A., Paleka, D., Panickssery, N., Gurnee, W., and Nanda, N. Refusal in language models is mediated by a single direction. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2024. arXiv preprint arXiv:2406.11717.
- Bailey, L., Serrano, A., Sheshadri, A., Seleznyov, M., Taylor, J., Jenner, E., Hilton, J., Casper, S., Guestrin, C., and Emmons, S. Obfuscated activations bypass LLM latent-space defenses. In *International Conference on Learning Representations (ICLR)*, 2025. arXiv preprint arXiv:2412.09565.
- Belrose, N., Ostrovsky, I., McKinney, L., Furman, Z., Smith, L., Halawi, D., Biderman, S., and Steinhardt, J. Eliciting latent predictions from transformers with the tuned lens. *arXiv preprint arXiv:2303.08112*, 2023.
- Gu, Q., Ju, Y., Sun, S., Gilitschenski, I., Nishimura, H., Itkina, M., and Shkurti, F. SAFE: Multitask failure detection for vision-language-action models. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2025. arXiv preprint arXiv:2506.09937.
- Häon, B., Stocking, K., Chuang, I., and Tomlin, C. Mechanistic interpretability for steering vision-language-action models. In *Conference on Robot Learning (CoRL)*, 2025. arXiv preprint arXiv:2509.00328.
- Hung, C.-Y., Majumder, N., Deng, H., Renhang, L., Ang, Y., Zadeh, A., Li, C., Herremans, D., Wang, Z., and Poria, S. NORA-1.5: A vision-language-action model trained using world model- and action-based preference rewards. *arXiv preprint arXiv:2511.14659*, 2025.
- Kim, M. J., Pertsch, K., Karamcheti, S., Xiao, T., Balakrishna, A., Nair, S., Rafailov, R., Foster, E., Lam, G., Sanketi, P., Vuong, Q., Kollar, T., Burchfiel, B., Tedrake, R., Sadigh, D., Levine, S., Liang, P., and Finn, C. Openvla: An open-source vision-language-action model. *arXiv preprint arXiv:2406.09246*, 2024.
- Lee, B. W., Padhi, I., Ramamurthy, K. N., Miehl, E., Dognin, P., Nagireddy, M., and Dhurandhar, A. Programming refusal with conditional activation steering. In *International Conference on Learning Representations (ICLR)*, 2025. arXiv preprint arXiv:2409.05907.
- Li, Q., Yin, B., Huang, W., Liu, R., Zou, B., Yu, R., Ye, J., Yu, W., and Wang, X. Vision-language-action safety: Threats, challenges, evaluations, and mechanisms. *arXiv preprint arXiv:2604.23775*, 2026.
- Liu, B., Zhu, Y., Gao, C., Feng, Y., Liu, Q., Zhu, Y., and Stone, P. LIBERO: Benchmarking knowledge transfer for lifelong robot learning. In *NeurIPS Datasets and Benchmarks Track*, 2023.
- Lu, H., Li, H., Shahani, P. S., Herbers, S., and Scheutz, M. Probing a vision-language-action model for symbolic states and integration into a cognitive architecture. *arXiv preprint arXiv:2502.04558*, 2025.
- Ravichandran, Z., Robey, A., Kumar, V., Pappas, G. J., and Hassani, H. Safety guardrails for LLM-enabled robots. *arXiv preprint arXiv:2503.07885*, 2025.
- Ren, A. Z., Dixit, A., Bodrova, A., Singh, S., Tu, S., Brown, N., Xu, P., Takayama, L., Xia, F., Varley, J., Xu, Z., Sadigh, D., Zeng, A., and Majumdar, A. Robots that ask for help: Uncertainty alignment for large language model planners. In *Conference on Robot Learning (CoRL)*, 2023. arXiv preprint arXiv:2307.01928.
- Robey, A., Ravichandran, Z., Kumar, V., Hassani, H., and Pappas, G. J. Jailbreaking LLM-controlled robots. In *IEEE International Conference on Robotics and Automation (ICRA)*, 2025. arXiv preprint arXiv:2410.13691.
- Turner, A. M., Thiergart, L., Leech, G., Udell, D., Vazquez, J. J., Mini, U., and MacDiarmid, M. Activation addition: Steering language models without optimization. *arXiv preprint arXiv:2308.10248*, 2023.
- Wang, C., Wei, Z., Liu, Q., and Chen, M. False sense of security: Why probing-based malicious input detection fails to generalize. *arXiv preprint arXiv:2509.03888*, 2025.
- Wen, S., Yang, S., Fu, S., Zhang, J., Hu, L., and Wang, D. Concept-based dictionary learning for inference-time safety in vision language action models. *arXiv preprint arXiv:2602.01834*, 2026.
- Yin, S., Pang, X., Ding, Y., Chen, M., Bi, Y., Xiong, Y., Huang, W., Xiang, Z., Shao, J., and Chen, S. SafeAgent-Bench: A benchmark for safe task planning of embodied LLM agents. *arXiv preprint arXiv:2412.13178*, 2024.
- Ying, Z., Wang, L., Xiao, Y., Wang, J., Ma, Y., Guo, J., Yin, Z., Zhang, M., Liu, A., and Liu, X. AGENTS SAFE: Benchmarking the safety of embodied agents on hazardous instructions. *arXiv preprint arXiv:2506.14697*, 2025.
- Zawalski, M., Chen, W., Pertsch, K., Mees, O., Finn, C., and Levine, S. Robotic control via embodied chain-of-thought reasoning. *arXiv preprint arXiv:2407.08693*, 2024.
- Zhang, B., Zhang, Y., Ji, J., Lei, Y., Cai, Y., Dai, J., Chen, Y., and Yang, Y. SafeVLA: Towards safety alignment of vision-language-action model via constrained learning. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2025a. arXiv preprint arXiv:2503.03480.

Zhang, H., Zhu, C., Wang, X., Zhou, Z., Yin, C., Li, M.,
Xue, L., Wang, Y., Hu, S., Liu, A., Guo, P., and Zhang,
L. Y. BadRobot: Jailbreaking embodied LLM agents
in the physical world. In *International Conference on
Learning Representations (ICLR)*, 2025b. arXiv preprint
arXiv:2407.20242.

Zhao, J., Huang, J., Wu, Z., Bau, D., and Shi, W. LLMs
encode harmfulness and refusal separately. In *Advances
in Neural Information Processing Systems (NeurIPS)*,
2025. arXiv preprint arXiv:2507.11878.

Zou, A., Phan, L., Chen, S., Campbell, J., Guo, P., Ren, R.,
Pan, A., Yin, X., Mazeika, M., Dombrowski, A.-K., et al.
Representation engineering: A top-down approach to AI
transparency. *arXiv preprint arXiv:2310.01405*, 2023.